Table 6-8. Visual Basic Serial Interface Program

```
Public gSend As Boolean                                'Global used for Send button state

Private Sub cmdSend_Click()                            'Routine to handle Send button press
   gSend = True                                        'Set Flag to True
End Sub

Private Sub Form_Load()                                'Main code section
   Dim strReturn As String                             'Used to return response
   Dim strHold As String                               'Temporary character space
   Dim Term As String                                  'Terminators
   Dim ZeroCount As Integer                            'Counter used for Timing out
   Dim strCommand As String                            'Data string sent to instrument

   frmSerial.Show                                      'Show main window
   Term = Chr(13) & Chr(10)                            'Terminators are <CR><LF>
   ZeroCount = 0                                       'Initialize counter
   strReturn = ""                                      'Clear return string
   strHold = ""                                        'Clear holding string
   If frmSerial.MSComm1.PortOpen = True Then           'Close serial port to change settings
       frmSerial.MSComm1.PortOpen = False
   End If
   frmSerial.MSComm1.CommPort = 1                      'Example of Comm 1
   frmSerial.MSComm1.Settings = "9600,o,7,1"           'Example of 9600 Baud,Parity,Data,Stop
   frmSerial.MSComm1.InputLen = 1                      'Read one character at a time
   frmSerial.MSComm1.PortOpen = True                   'Open port

 Do
   Do                                                  'Wait loop
   DoEvents                                            'Give up processor to other events
   Loop Until gSend = True                             'Loop until Send button pressed
   gSend = False                                       'Set Flag as false

   strCommand = frmSerial.txtCommand.Text              'Get Command
   strReturn = ""                                      'Clear response display

   strCommand = UCase(strCommand)                      'Set all characters to upper case
   If strCommand = "EXIT" Then                         'Get out on EXIT
       End
   End If

   frmSerial.MSComm1.Output = strCommand & Term        'Send command to instrument
   If InStr(strCommand, "?") <> 0 Then                 'Check to see if query
       While (ZeroCount < 20) And (strHold <> Chr$(10))  'Wait for response
           If frmSerial.MSComm1.InBufferCount = 0 Then   'Add 1 to timeout if no character
               frmSerial.Timer1.Enabled = True
               Do
               DoEvents                                'Wait for 10 millisecond timer
               Loop Until frmSerial.Timer1.Enabled = False
               ZeroCount = ZeroCount + 1               'Timeout at 2 seconds
           Else
               ZeroCount = 0                           'Reset timeout for each character
               strHold = frmSerial.MSComm1.Input       'Read in one character
               strReturn = strReturn + strHold         'Add next character to string
           End If
       Wend                                            'Get characters until terminators

       If strReturn <> "" Then                         'Check if string empty
           strReturn = Mid(strReturn, 1, InStr(strReturn, Term) - 1)  'Strip terminators
       Else
           strReturn = "No Response"                   'Send No Response
       End If
       frmSerial.txtResponse.Text = strReturn          'Put response in textbox on main form
       strHold = ""                                    'Reset holding string
       ZeroCount = 0                                   'Reset timeout counter
   End If
 Loop
End Sub

Private Sub Timer1_Timer()                             'Routine to handle Timer interrupt
   frmSerial.Timer1.Enabled = False                    'Turn off timer
End Sub
```

6.2.7.2 Program Operation

Once either example program is running, try the following commands and observe the response of the instrument. Input from the user is shown in **bold** and terminators are added by the program. The word [term] indicates the required terminators included with the response.

`ENTER COMMAND?` **`*IDN?`**	Identification query. Instrument will return a string identifying itself.
`RESPONSE: LSCI,MODEL475,1234567,02032003[term]`	
`ENTER COMMAND?` **`RDGFIELD?`**	Field reading query. Instrument will return a string with the present field reading in the present units.
`RESPONSE: +273.150E+00[term]`	
`ENTER COMMAND?` **`RANGE 1`**	Field range command. Instrument will set to the lowest range. No response will be sent.
`ENTER COMMAND?` **`RANGE?`**	Field range query. Instrument will return a string with the present field range setting.
`RESPONSE: 1[term]`	
`ENTER COMMAND?` **`RANGE 5;RANGE?`**	Field range command followed by a query. Instrument will change to the highest range setting then return a string
`RESPONSE: 5[term]`	with the present setting.

The following are additional notes on using either Serial Interface program.

- If you enter a correctly spelled query without a "**?**," nothing will be returned. Incorrectly spelled commands and queries are ignored. Commands and queries should have a space separating the command and associated parameters.
- Leading zeros and zeros following a decimal point are not needed in a command string, but are sent in response to a query. A leading "+" is not required but a leading "–" *is* required.

6.2.8 Troubleshooting

New Installation

1. Check instrument Baud rate.
2. Make sure transmit (TD) signal line from the instrument is routed to receive (RD) on the computer and vice versa. (Use a null modem adapter if not).
3. Always send terminators.
4. Send entire message string at one time including terminators. (Many terminal emulation programs do not.)
5. Send only one simple command at a time until communication is established.
6. Be sure to spell commands correctly and use proper syntax.

Old Installation No Longer Working

7. Power instrument off then on again to see if it is a soft failure.
8. Power computer off then on again to see if communication port is locked up.
9. Verify that Baud rate has not been changed on the instrument during a memory reset.
10. Check all cable connections.

Intermittent Lockups

11. Check cable connections and length.
12. Increase delay between all commands to 100 ms to make sure instrument is not being over loaded.

6.3 COMMAND SUMMARY

This paragraph provides a listing of the IEEE-488 and Serial Interface Commands. A summary of all the commands is provided in Table 6-9. All the commands are detailed in Paragraph 6.3.1, presented in alphabetical order.

Sample Command Format

IEEE IEEE-488 Interface Parameter Command

Input: **IEEE <terminator>, <EOI enable>, <address>[term]**

Format: n,n,nn

<terminator> Specifies the terminator. Valid entries: 0 = <CR><LF>,1 = <LF><CR>, 2 = <LF>, 3 = no terminator (must have EOI enabled).

<EOI enable> Sets EOI mode: 0 = enabled, 1 = disabled.

<address> Specifies the IEEE address: 1 – 30. (Address 0 and 31 are reserved.)

Example: **IEEE 0,0,4[term]** – After receipt of the current terminator, the instrument uses EOI mode, uses <CR><LF> as the new terminator, and responds to address 4.

Sample Query Format

IEEE? IEEE-488 Interface Parameter Query

Input: **IEEE?[term]**

Returned: <terminator>, <EOI enable>, <address>[term]

Format: n,n,nn (Refer to command for description)

Key

∗	Begins common interface command.
?	Required to identify queries.
aa…	String of alpha numeric characters.
±nnn.nnnE±nn	Number represented in scientific notation format.
[term]	Terminator characters.
<…>	Indicated a parameter field, many are command specific.
<state>	Parameter field with only On/Off states.

NOTE: Any number being represented in scientific notation may also be entered as a string of number characters. The following example shows two different ways of sending the same command. Refer to the individual command descriptions for further details.

RELSP 2.0E+03[term] Command will set the Relative setpoint value to 2000.

RELSP 2000[term] Command will set the Relative setpoint value to 2000.

Table 6-9. Command Summary

Command	Function	Page
*CLS	Clear Interface Cmd	6-27
*ESE	Standard Event Status Enable Register Cmd	6-27
*ESE?	Standard Event Status Enable Register Query	6-27
*ESR?	Standard Event Status Register Query	6-27
*IDN?	Identification Query	6-27
*OPC	Operation Complete Cmd	6-27
*OPC?	Operation Complete Query	6-28
*RST	Reset Instrument Cmd	6-28
*SRE	Service Request Enable Register Cmd	6-28
*SRE?	Service Request Enable Register Query	6-28
*STB?	Status Byte Query	6-28
*TRG	Trigger Event	6-28
*TST?	Self-Test Query	6-28
*WAI	Wait-To-Continue Cmd	6-28
ALARM	Alarm Parameter Cmd	6-29
ALARM?	Alarm Parameter Query	6-29
ALARMST?	Alarm Status Query	6-29
ANALOG	Analog Output 3 Parameter Cmd	6-29
ANALOG?	Analog Output 3 Parameter Query	6-30
AOUT?	Analog Output 3 Data Query	6-30
AUTO	Auto Range Command	6-30
AUTO?	Auto Range Query	6-30
BAUD	RS-232 Baud Rate Cmd	6-30
BAUD?	RS-232 Baud Rate Query	6-30
BEEP	Audible Alarm Beeper Cmd	6-30
BEEP?	Audible Alarm Beeper Query	6-30
BRIGT	Display Brightness Cmd	6-30
BRIGT?	Display Brightness Query	6-31
CMODE	Field Control Mode Cmd	6-31
CMODE?	Field Control Mode Query	6-31
CPARAM	Field Control Parameter Cmd	6-31
CPARAM?	Field Control Parameter Query	6-31
CSETP	Field Control Setpoint Cmd	6-31
CSETP?	Field Control Setpoint Query	6-31
DFLT	Factory Defaults Cmd	6-32
DISPLAY	Display Configuration Cmd	6-32
DISPLAY?	Display Configuration Query	6-32
DLOG	Datalog Start Cmd	6-32
DLOGNUM?	Datalog Number of Points Query	6-32
DLOGRDG?	Datalog Reading Query	6-32
DLOGSET	Datalog Rate Cmd	6-33
DLOGSET?	Datalog Rate Query	6-33
FILTER	Filter Parameter Cmd	6-33
FILTER?	Filter Parameter Query	6-33
IEEE	IEEE-488 Interface Parameter Cmd	6-33
IEEE?	IEEE-488 Interface Parameter Query	6-34
KEYST?	Keypad Status Query	6-34
LOCK	Front Panel Keyboard Lock Cmd	6-34
LOCK?	Front Panel Keyboard Lock Query	6-34
MXHOLD	Max Hold Cmd	6-34
MXHOLD?	Max Hold Query	6-34
MXRST	Max Hold Reset Cmd	6-34
MODE	Remote Interface Mode Cmd	6-35
MODE?	Remote Interface Mode Query	6-35
OPST?	Operational Status Query	6-35
OPSTE	Operational Status Enable Cmd	6-35
OPSTE?	Operational Status Enable Query	6-35
OPSTR?	Operational Status Register Query	6-35
PKRST	Peak Hold Reset Cmd	6-35
PRBFCOMP	Probe Field Compensation Cmd	6-36
PRBFCOMP?	Probe Field Compensation Query	6-36
PRBSENS?	Probe Sensitivity Query	6-36
PRBSNUM?	Probe Serial Number Query	6-36
PRBTCOMP	Probe Temperature Compensation Cmd	6-36
PRBTCOMP?	Probe Temperature Compensation Query	6-36
RAMPST?	Field Control Setpoint Ramp Status Query	6-36
RANGE	Field Range Cmd	6-36
RANGE?	Field Range Query	6-37
RDGFAST?	Binary Field Reading Query	6-37
RDGFIELD?	Field Reading Query	6-37
RDGMODE	Measurement Mode Cmd	6-37
RDGMODE?	Measurement Mode Query	6-37
RDGFRQ?	Frequency Reading Query	6-37
RDGMNMX?	Minimum and Maximum Reading Query	6-38
RDGOHM?	Resistance Reading Query	6-38
RDGPEAK?	Peak Reading Query	6-38
RDGREL?	Relative Reading Query	6-38
RDGTEMP?	Probe Temperature Reading Query	6-38
REL	Relative Mode Cmd	6-38
REL?	Relative Mode Query	6-38
RELAY	Relay Parameter Cmd	6-39
RELAY?	Relay Parameter Query	6-39
RELAYST?	Relay Status Query	6-39
RELSP	Relative Setpoint Cmd	6-39
RELSP?	Relative Setpoint Query	6-39
TRIG	Hardware Trigger Out Cmd	6-39
TRIG?	Hardware Trigger Out Query	6-39
TUNIT	Probe Temperature Units Cmd	6-40
TUNIT?	Probe Temperature Units Query	6-40
TYPE?	Probe Type Query	6-40
UNIT	Field Units Cmd	6-40
UNIT?	Field Units Query	6-40
ZCLEAR	Clear Zero Probe Cmd	6-40
ZPROBE	Zero Probe Cmd	6-40

6.3.1 Interface Commands (Alphabetical Listing)

***CLS** Clear Interface Command

Input: `*CLS [term]`

Remarks: Clears the bits in the Standard Event Status Register and Operation Event Register and terminates all pending operations. Clears the interface, but *not* the instrument. The related instrument command is ***RST**.

***ESE** Standard Event Status Enable Register Command

Input: `*ESE <bit weighting>[term]`

Format: nnn

Remarks: The Standard Event Status Enable Register determines which bits in the Standard Event Status Register will set the summary bit in the Status Byte. This command programs the enable register using a decimal value which corresponds to the binary-weighted sum of all bits in the register. Refer to Paragraph 6.1.4.2.1.

***ESE?** Standard Event Status Enable Register Query

Input: `*ESE? [term]`

Returned: <bit weighting>[term]

Format: nnn (Refer to command for description)

***ESR?** Standard Event Status Register Query

Input: `*ESR? [term]`

Returned: <bit weighting>

Format: nnn

Remarks: Bits in this register correspond to various system events and latch when the event occurs. When an event bit is set, subsequent events corresponding to that bit are ignored. Set bits remain latched until the register is reset by this query or a *CLS command. Refer to Paragraph 6.1.4.2.1.

***IDN?** Identification Query

Input: `*IDN? [term]`

Returned: <manufacturer>,<model>,<serial>,<date>[term]

Format: aaaa,aaaaaaaa,aaaaaaa,mmddyyyy

<manufacture>	Manufacturer ID
<model>	Instrument model number
<serial>	Serial number
<date>	Instrument firmware revision date

Example: **LSCI,MODEL475,1234567,06122003**

***OPC** Operation Complete Command

Input: `*OPC [term]`

Remarks: Used in conjunction with bit 0 (OPC of the Standard Event Status Register. If sent as the last command in a command sequence, bit 0 will be set when the instrument completes the operation that was initiated by the command sequence. Refer to Paragraph 6.1.4.3.6 for more information.

***OPC?** Operation Complete Query

Input: `*OPC? [term]`

Returned: 1[term]

Remarks: Has *no* interaction with bit 0 (OPC) of the Standard Event Status Register. If sent at the end of a command sequence, the bus will be held until the instrument completes the operation that was initiated by the command sequence. Once the sequence is complete a 1 will be placed in the output buffer. Refer to Paragraph 6.1.4.3.6 for more information.

***RST** Reset Instrument Command

Input: `*RST[term]`

Remarks: Sets controller parameters to power-up settings. Use the **DFLT** command to set factory defaults.

***SRE** Service Request Enable Register Command

Input: `*SRE <bit weighting>[term]`

Format: nnn

Remarks: The Service Request Enable Register determines which summary bits of the Status Byte may set bit 6 (RQS/MSS) of the Status Byte to generate a Service Request. This command programs the enable register using a decimal value which corresponds to the binary-weighted sum of all bits in the register. Refer to Paragraph 6.1.4.3

***SRE?** Service Request Enable Register Query

Input: `*SRE? [term]`

Returned: <bit weighting>[term]

Format: nnn (Refer to command for description)

***STB?** Status Byte Query

Input: `*STB? [term]`

Returned: <bit weighting>[term]

Format: nnn

Remarks: This command is similar to a Serial Poll except it is processed like any other instrument command. It returns the same result as a Serial Poll except that the Status Byte bit 6 (RQS/MSS) is not cleared. Refer to Paragraph 6.1.4.3.4

***TRG** Trigger Event

Input: `*TRG [term]`

Remarks: Starts the datalog capture mode. This command is equivalent in operation to the DLOG command.

***TST?** Self-Test Query

Input: `*TST? [term]`

Returned: <status>[term]

Format: n

<status> 0 = No errors found, 1 = Errors found

Remarks: The Model 475 reports status based on test done at power up.

***WAI** Wait-to-Continue Command

Input: `*WAI [term]`

Remarks: This command is not supported in the Model 475.

ALARM

Alarm Parameter Command

Input: **ALARM <off/on>, <mode>, <low value>, <high value>, <out/in>, <sort>[term]**

Format: n,n,±nnn.nnnE±nn,±nnn.nnnE±nn,n,n

<off/on>	Specifies alarm checking on or off: 0 = Off, 1 = On.
<mode>	Specifies checking magnitude(absolute value used) or algebraically(includes sign): 1 = Magnitude check, 2 = Algebraic check.
<low value>	Sets the value the source is checked against to activate the low alarm: ±350 kG
<high value>	Sets the value the source is checked against to activate high alarm: ±350 kG
<out/in>	Specifies the alarm to trigger on value outside or inside of setpoints: 1= Outside, 2 = Inside.
<sort>	Specifies sort mode pass/fail checking off or on: 0 = Off, 1 = On.

Examples: **ALARM 1,1,100,300,1,0[term] –** Turns alarm checking on. Activates alarm if the absolute value of the field is over 300 Gauss, or if the absolute value of the field is below 100 Gauss. Sort mode is off.

ALARM 1,2,-100,300,2,0[term] – Turns alarm checking on. Activates alarm if the value of the field is between –100 Gauss and +300 Gauss. Sort mode is off.

ALARM?

Alarm Parameter Query

Input: **ALARM? [term]**

Returned: <off/on>, <mode>, <low value>, <high value>,<out/in>, <sort> [term]

Format: n,n,±nnn.nnnE±nn,±nnn.nnnE±nn,n, n (Refer to command for description)

ALARMST?

Alarm Status Query

Input: **ALARMST? [term]**

Returned: <state> [term]

Format: n

< state> 0 = Off, 1 = On

ANALOG

Analog Output 3 Parameter Command

Input: **ANALOG <mode>, <polarity>, <low value>, <high value>, <manual value>, <voltage limit> [term]**

Format: n,n,±nnn.nnnE±nn,±nnn.nnnE±nn, ±nnn.nnnE±nn ,nn

<mode>	Specifies data the analog out 3 monitors: 0 = off, 1 = default, 2 = user defined, 3 = manual, 4 = control
<polarity>	Specifies analog output polarity: 1 = unipolar, 2 = bipolar.
<low value>	If <mode> is 2, this parameter represents the data at which the analog output reaches –100% output if bipolar, or 0% output if unipolar: ±350 kG
<high value>	If <mode> is 2, this parameter represents the data at which the analog output reaches +100% output: ±350 kG
<manual value>	If <mode> is 3, this parameter represents the percent output of the analog output between –100% and +100%.
<voltage limit>	Specifies absolute maximum analog output voltage: 1 to 10 V

Remarks: If the analog mode changes from control to manual (4 to 3), the current analog output will be stored to the manual value. This ensures a smooth transition from control mode to manual mode.

Example: **ANALOG 2,2,25.000E-3,250.000E-3,0.00000E+00,10[term]** – Configures Analog Output 3 for user defined mode, bipolar polarity, –100% out at 25 mG, +100% out at 250 mG, and an absolute maximum output voltage of 10 V. The manual value is ignored in this mode.

ANALOG? Analog Output 3 Parameter Query

Input: `ANALOG? [term]`

Returned: <mode>, <polarity>, <low value>, <high value>, <manual value>, <voltage limit> [term]

Format: n,n,±nnn.nnnE±nn,±nnn.nnnE±nn,±nnn.nnnE±nn (Refer to command for definition)

AOUT? Analog Output 3 Data Query

Input: `AOUT? [term]`

Returned: <percent>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the percentage of output of Analog Out 3.

AUTO Auto Range Command

Input: `AUTO <off/on>[term]`

Format: n

<off/on> Specifies autorange on or off: 0 = Off, 1 = On.

Example: **AUTO 1 [term]** – Turns on the autorange feature.

AUTO? Auto Range Query

Input: `AUTO? [term]`

Returned: <off/on>[term]

Format: n (Refer to command for description)

BAUD RS-232 Baud Rate Command

Input: `BAUD <bps>[term]`

Format: n

<bps> 1 = 9600 Baud, 2 = 19200 Baud, 3 = 38400 Baud, 4 = 57600 Baud.

BAUD? RS-232 Baud Rate Query

Input: `BAUD?[term]`

Returned: <bps>[term]

Format: n (Refer to command for description)

BEEP Alarm Beeper Command

Input: `BEEP <state>[term]`

Format: n

<state> 0 = Off, 1 = On.

Remarks: Enables or disables system beeper sound when an alarm condition is met.

BEEP? Alarm Beeper Query

Input: `BEEP? [term]`

Returned: <state>[term]

Format: n (Refer to command for description)

BRIGT Display Brightness Command

Input: `BRIGT <bright>[term]`

Format: n

<bright> 1 = 25%, 2 = 50%, 3 = 75%, 4 = 100%. Default = 3.

BRIGT? Display Brightness Query

Input: `BRIGT? [term]`

Returned: <bright>[term]

Format: n (Refer to command for description)

CMODE Field Control Mode Command

Input: `CMODE <mode>[term]`

Format: n

<mode> Specifies the control mode. Valid entries: 0 = Off, 1 = Closed Loop PI.

Example: **CMODE 1[term]** – Field Control uses Closed Loop control.

CMODE? Field Control Mode Query

Input: `CMODE? [term]`

Returned: <mode>[term]

Format: n (Refer to command for description)

CPARAM Field Control Parameter Command

Input: `CPARAM <P value>, <I value>, <ramp rate>, <control slope limit>[term]`

Format: +nnn.nnnE±nn,+nnn.nnnE±nn,+nnn.nnnE±nn,+nnn.nnnE±nn

<P value> The value for control loop Proportional (gain): 0.01 to 1000.

<I value> The value for control loop Integral (reset): 0.0001 to 1000

<ramp rate> Field Setpoint ramp in Units/minute, where Units are the present Field measurement units. Setting to 0 will turn ramping off.

<control slope limit> Analog Output limit in Volts/minute: 0.01 to 1000.

Example: **CPARAM 10,50,3000,75[term]** – Field Control P is 10, I is 50 and when the **CSETP** command is issued, the field setpoint will ramp at a rate of 3000 Gauss/minute. The Analog Output will be limited to a 75 Volts/minute change.

CPARAM? Field Control Parameter Query

Input: `CPARAM? [term]`

Returned: <P value>, <I value>, <ramp rate>, <control slope limit>

Format: ±nnnnnn,±nnnnnn,±nnn.nnnE±nn,±nnn.nnnE±nn (Refer to command for description)

CSETP Field Control Setpoint Command

Input: `CSETP <value>[term]`

Format: ±nnn.nnnE±nn

<value> Specifies the final setpoint of the Field Control Ramp: ±350 kG

Example: **CSETP 3000[term]** – Ramp the field setpoint to 3000 (resent units) based on parameters configured in the command **CPARAM**.

CSETP? Field Control Present Setpoint Query

Input: `CSETP? [term]`

Returned: <setpoint> [term]

Format: ±nnn.nnnE±nn (Refer to command for description)

Remarks: Returns the present field setpoint. If the setpoint is ramping this will be the value where the ramp is, not the final destination setting.

DFLT Factory Defaults Command

Input: `DFLT 99 [term]`

Remarks: Sets all configuration values to factory defaults and resets the instrument. The "99" is included to prevent accidentally setting the unit to defaults.

DISPLAY Display Configuration Command

Input: `DISPLAY <item>[term]`

Format: n

<item> Specifies the item to display on line 2: 1 – 6

1 = Blank
2 = Field
3 = Probe Temperature and Frequency (if available)
4 = Relative setpoint
5 = Field Control setpoint
6 = Alarm Sort message

Example: **DISPLAY 3[term]** – Probe Temperature will be displayed on Line 2 if the probe is capable of measuring temperature. The Frequency will also be displayed if the Model 475 is in RMS mode.

DISPLAY? Display Configuration Query

Input: `DISPLAY? [term]`

Returned: <item>[term]

Format: n (Refer to command for description)

DLOG Datalog Stop/Start Command

Input: `DLOG <stop/start> [term]`

Format: n

<stop/start> Starts and stops the datalog function: 0 = Stop, 1 = Start.

Remarks: Use the DLOGSET command to set the datalog rate. The DLOGNUM? command can be used to determine how many datalog points have been acquired.

DLOGNUM? Datalog Number of Points Query

Input: `DLOGNUM? [term]`

Returned: <num rdgs>[term]

Format: nnnn

Remarks: Returns the number of field readings stored in the Datalog buffer.

DLOGRDG? Datalog Reading Query

Input: `DLOGRDG? <data point>[term]`

Format: nnnn

<data point> Specifies data point to retrieve: 1 – 1024.

Returned: <value>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the field reading stored in the Datalog buffer at the point specified. Readings are in the field units the instrument is currently set to.

DLOGSET Datalog Rate Command

Input: `DLOGSET <rate> [term]`

Format: n

<rate> Configures the Datalog readings per second:
- 1 = 1 reading per second
- 2 = 10 readings per second
- 3 = 30 readings per second
- 4 = 100 readings per second
- 5 = 200 readings per second
- 6 = 400 readings per second
- 7 = 800 readings per second
- 8 = 1000 readings per second

DLOGSET? Datalog Rate Query

Input: `DLOGSET? [term]`

Returned: <rate> [term]

Format: n (Refer to command for description)

FILTER Filter Parameter command

Input: `FILTER <rmsbpw>, <rmsbpn>, <rmslp>[term]`

Format: n,n,n

<rmsbpw> Selects center frequency of the RMS wide band filter: 1 = 100 Hz, 2 = 200 Hz, 3 = 400 Hz, 4 = 800 Hz, 5 = 1600 Hz, 6 = 3200 Hz, 7 = 6400 Hz, 8 = no filter.

<rmsbpn> Selects center frequency of the RMS narrow band filter: 1 = 50 Hz, 2 = 60 Hz, 3 = 100 Hz, 4 = 120 Hz, 5 = 200 Hz, 6 = 400 Hz, 7 = 800 Hz.

<rmslp> Selects corner frequency of the RMS low pass filter: 1 = 20 Hz, 2 = 60 Hz, 3 = 140 Hz.

Example: **FILTER 3,5,2[term]** – Configures the RMS wide band filter center frequency to 400 Hz, the RMS narrow band center frequency to 200 Hz, and the RMS low pass corner frequency to 60 Hz.

FILTER? Filter Parameter Query

Input: `FILTER? [term]`

Returned: <rmsbpw>, <rmsbpn>, <rmslp>[term]

Format: n,n,n (Refer to command for description)

IEEE IEEE-488 Interface Parameter Command

Input: `IEEE <terminator>, <EOI enable>, <address>[term]`

Format: n,n,nn

<terminator> Specifies the terminator. Valid entries: 0 = <CR><LF>,1 = <LF><CR>, 2 = <LF>, 3 = no terminator (must have EOI enabled).

<EOI enable> Sets EOI mode: 0 = enabled, 1 = disabled.

<address> Specifies the IEEE address: 1 – 30. (Address 0 and 31 are reserved.)

Example: **IEEE 0,0,4[term]** – After receipt of the current terminator, the instrument uses EOI mode, uses <CR><LF> as the new terminator, and responds to address 4.

IEEE? IEEE-488 Interface Parameter Query

Input: `IEEE? [term]`

Returned: <terminator>, <EOI enable>, <address>[term]

Format: n,n,nn (Refer to command for description)

KEYST? Keypad Status Query

Input: `KEYST?[term]`

Returned: <keypad status>[term]

Format: n

Remarks: Returns a number descriptor of the last key pressed since the last KEYST?. KEYST? returns 1 after initial power-up. Returns a 0 if no key pressed since last query.

LOCK Front Panel Keyboard Lock Command

Input: `LOCK <state>, <code>[term]`

Format: n,nnn

<state> 0 = Unlocked, 1 = Locked

<code> Specifies lock-out code. Valid entries are 000 – 999.

Remarks: Locks out all front panel entries.

Example: **LOCK 1,123[term] –** Enables keypad lock and sets the code to 123.

LOCK? Front Panel Keyboard Lock Query

Input: `LOCK?[term]`

Returned: <state>, <code>[term]

Format: n,nnn (Refer to command for description)

MXHOLD Max Hold Command

Input: `MXHOLD <off/on>,<mode>,<display>[term]`

Format: n,n,n

<off/on> Specifies Max Hold on or off: 0 = off and 1 = on.

<mode> Specifies checking magnitude(absolute value used) or algebraically(includes sign): 1 = Magnitude check, 2 = Algebraic check.

<display> Specifies Display configuration when Max Hold is on: 1 = Display Maximum value on top line, 2 = Display Minimum value on top line, 3 = Display Both, Max on top line and Minimum on bottom line. This overrides the DISPLAY command configuration.

Example: **MXHOLD 1,1,2[term] –** Turns the Max Hold feature on using the Magnitude checking mode. The Model 475 displays both Max and Min values.

MXHOLD? Max Hold Query

Input: `MODE? [term]`

Returned: <off/on>,<mode>,<display> [term]

Format: n,n,n (Refer to command for description)

MXRST Max Hold Reset Command

Input: `MXRST [term]`

Remarks: Resets the minimum and maximum stored field readings and sets them equal to the present reading.

MODE Remote Interface Mode Command

Input: `MODE <mode>[term]`

Format: n

<mode> 0 = local, 1 = remote, 2 = remote with local lockout.

Example: **MODE 2[term] –** Places the Model 475 into remote mode with local lockout.

MODE? Remote Interface Mode Query

Input: `MODE? [term]`

Returned: <mode>[term]

Format: n (Refer to command for description)

OPST? Operational Status Query

Input: `OPST? [term]`

Returned: <bit weighting> [term]

Format: nnn

Remarks: The integer returned represents the sum of the bit weighting of the operational status bits. Refer to Paragraph 6.1.4.2.2 for a list of operational status bits.

OPSTE Operational Status Enable Command

Input: `OPSTE <bit weighting>[term]`

Format: nnn

Remarks: Each bit has a bit weighting and represents the enable/disable mask of the corresponding operational status bit in the Operational Status Register. This determines which status bits can set the corresponding summary bit in the Status Byte Register. To enable a status bit, send the command **OPSTE** with the sum of the bit weighting for each desired bit. Refer to Paragraph 6.1.4.2.2 for a list of operational status bits.

OPSTE? Operational Status Enable Query

Input: `OPSTE? [term]`

Returned: <bit weighting> [term]

Format: nnn Refer to Paragraph 6.1.4.2.2 for a list of operational status bits.

OPSTR? Operational Status Register Query

Input: `OPSTR? [term]`

Returned: <bit weighting> [term]

Format: nnn

Remarks: The integers returned represent the sum of the bit weighting of the operational status bits. These status bits are latched when the condition is detected. This register is cleared when it is read. Refer to Paragraph 6.1.4.2.2 for a list of operational status bits.

PKRST Peak Hold Reset Command

Input: `PKRST [term]`

Remarks: Resets the stored positive and negative peak field readings and sets them equal to zero.
This is only valid in Pulse measurement mode.

PRBFCOMP Probe Field Compensation Command

Input: `PRBFCOMP <off/on>[term]`

Format: n

<off/on> Specifies Probe Field compensation off or on. Valid entries: 0 = off, 1 = on.

Example: **PRBFCOMP 1[term]** – Field Measurement uses the Probe Field Compensation table

PRBFCOMP? Probe Field Compensation Query

Input: `PRBFCOMP? [term]`

Returned: <off/on>[term]

Format: n (Refer to command for description)

PRBSENS? Probe Sensitivity Query

Input: `PRBSENS? [term]`

Returned: <sensitivity>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the probe sensitivity in mV/kG.

PRBSNUM? Probe Serial Number Query

Input: `PRBSNUM? [term]`

Returned: <type>[term]

Format: xxxxxxxxxx

Remarks: Returns the probe serial number.

PRBTCOMP Probe Temperature Compensation Command

Input: `PRBTCOMP <off/on>[term]`

Format: n

<off/on> Specifies Probe Temperature compensation off or on. Valid entries: 0 = off, 1 = on.

Example: **PRBTCOMP 1[term]** – Field Measurement is compensated for present probe temperature.

PRBTCOMP? Probe Temperature Compensation Query

Input: `PRBTCOMP? [term]`

Returned: <off/on>[term]

Format: n (Refer to command for description)

RAMPST? Field Control Setpoint Ramp Status Query

Input: `RAMPST? [term]`

Returned: <ramp status>[term]

Format: n

<ramp status> 0 = Not ramping, 1 = Field control setpoint is ramping.

RANGE Field Range Command

Input: `RANGE <range>[term]`

Format: n

<range> Specifies range from lowest to highest: 1 – 5. (Field values are probe dependent.)

Example: RANGE 4 [term] – Sets the present range to 4.

RANGE? Field Range Query

Input: `RANGE? [term]`

Returned: <range>[term]

Format: n (Refer to command for description)

RDGFAST? Binary Field Reading Query

Input: `RDGFAST?<num rdgs>[term]`

Format: nnnnn

<num rdgs> Specifies number of readings to retrieve: 1 – 99999.

Returned: <field>[term]

Format: Refer to Paragraph 6.1.3.5 for IEEE-754 format description.

Remarks: Returns a block of binary field reading data.

RDGFIELD? Field Reading Query

Input: `RDGFIELD? [term]`

Returned: <field>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the field reading in a format based on the present units. This is valid for DC or RMS.

RDGMODE Measurement Mode Command

Input: `RDGMODE <mode>, <dc resolution>, <rms filter mode>, <peak mode>, <peak disp>[term]`

Format: n,n,n,n,n

<mode>	Specifies the measurement mode: 1 = DC, 2 = RMS, 3 = peak.
<dc resolution>	DC operating resolution in number of digits: 1 = 3 digits, 2 = 4 digits, 3 = 5 digits
<rms filter mode>	Filter band type for RMS measurement: 1 = wide band, 2 = narrow band, 3 = low pass.
<peak mode>	Specifies peak measurement mode: 1 = periodic, 2 = pulse.
<peak disp>	Specifies display of peak reading: 1 = positive, 2 = negative, 3 = both.

Remarks: Reference the **FILTER** command for information on configuring the filter parameters for each measurement mode.

Example: **RDGMODE 1,3,1,1,1[term]** – The Model 475 is configured for DC field measurement, dc resolution of 5 digits, wide band rms filter mode, peak measurement mode is periodic, and positive peak readings will be displayed if the measurement mode is changed to peak.

RDGMODE? Measurement Mode Query

Input: `RDGMODE? [term]`

Returned: <mode>, <dc resolution>, <rms filter mode>, <peak mode>, <peak disp>[term]

Format: n,n,n,n,n (Refer to command for description)

RDGFRQ? Frequency Reading Query

Input: `RDGFRQ? [term]`

Returned: <frequency>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the frequency reading in Hz. The instrument must be in RMS for this to be valid.

RDGMNMX? Minimum and Maximum Reading Query

Input: `RDGMNMX? [term]`

Returned: <min>,<max>[term]

Format: ±nnn.nnnE±nn, ±nnn.nnnE±nn

Remarks: Returns the most recent minimum and maximum field readings.

RDGOHM? Resistance Reading Query

Input: `RDGOHM? [term]`

Returned: <hall resistance>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the Hall resistance of the sensor.

RDGPEAK? Peak Reading Query

Input: `RDGPEAK? [term]`

Returned: <negative peak>,<positive peak> [term]

Format: ±nnn.nnnE±nn, ±nnn.nnnE±nn

Remarks: Returns the negative and positive peak readings.

RDGREL? Relative Reading Query

Input: `RDGREL? [term]`

Returned: <relative reading>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the relative field reading.

RDGTEMP? Probe Temperature Reading Query

Input: `RDGTEMP? [term]`

Returned: <temperature>[term]

Format: ±nnn.nnnE±nn

Remarks: Returns the probe temperature reading in a format based on the present temperature units.

REL Relative Mode Command

Input: `REL <off/on>, <setpoint source>[term]`

Format: n,n

<off/on> Specifies Relative mode off or on: 0 = off, 1 = on.

<setpoint source> Specifies source of relative setpoint: 1 = User defined, 2 = Present Field.

Example: **REL 1,1 [term]** – Relative mode turned on, configured to use the User defined setpoint.
Refer to RELSP command.

REL? Relative Mode Query

Input: `REL? [term]`

Returned: <off/on>, <setpoint source>[term]

Format: n,n (Refer to command for description)

RELAY Relay Parameter Command

Input: `RELAY <relay number>, <mode>, <alarm type>[term]`

Format: n,n,nn,n

<relay number> Specifies which relay to configure: 1 = Relay 1, 2 = Relay 2.

<mode> Specifies relay mode: 0 = Off, 1 = On, 2 = Alarms.

<alarm type> Specifies the alarm type that activates the relay when the relay is in alarm mode: 1 = Low Alarm, 2 = High Alarm, 3 = Both Alarms.

Example: **RELAY 1,2,2[term]** – Relay 1 is setup in Alarms mode and activates when the low alarm activates.

RELAY? Relay Parameter Query

Input: `RELAY? <relay number>[term]`

Format: n

<relay number> Specifies which relay to query: 1–2.

Returned: <mode>, <alarm type>[term]

Format: n,n (Refer to command for description)

RELAYST? Relay Status Query

Input: `RELAYST? <relay number>[term]`

Format: n

<relay number> Specifies which relay to query: 1 = Relay 1, 2 = Relay 2.

Returned: <status>[term]

Format: n 0 = Off, 1 = On.

RELSP Relative Setpoint Command

Input: `RELSP <setpoint>[term]`

Format: ±nnn.nnnE±nn

<setpoint> Specifies the setpoint to use in the relative calculation: ±350 kG

Example: **RELSP 1200[term]** – Configure the relative setpoint as 1200 Gauss(if units in Gauss). The relative reading will use this value if relative is using the user defined setpoint. Refer to REL command.

RELSP? Relative Setpoint Query

Input: `RELSP? [term]`

Returned: <setpoint>[term]

Format: ±nnn.nnnE±nn (Refer to command for description)

TRIG Hardware Trigger Out Command

Input: `TRIG <off/on>[term]`

Format: n

<off/on> Specifies hardware trigger off or on. Valid entries: 0 = Off, 1 = on.

Example: **TRIG 1[term]** – The hardware trigger line produces a trigger when the instrument calculates a new reading.

TRIG? Hardware Trigger Out Query

Input: `TRIG? [term]`

Returned: <off/on>[term]

Format: n (Refer to command for description)

TUNIT Probe Temperature Units Command

Input: `TUNIT <units>[term]`

Format: n

<units> 1 = Celsius, 2 = Kelvin

Example: **TUNIT 1[term] –** Configures the Model 475 to report probe temperature in °C.

TUNIT? Probe Temperature Units Query

Input: `TUNIT? [term]`

Returned: <units>[term]

Format: n (Refer to command for description)

TYPE? Probe Type Query

Input: `TYPE? [term]`

Returned: <type>[term]

Format: nn

Remarks: Returns the probe type:

40 = high sensitivity
41 = high stability
42 = ultra-high sensitivity
50 = user programmable cable/high sensitivity probe
51 = user programmable cable/high stability probe
52 = user programmable cable/ultra-high sensitivity probe

UNIT Field Units Command

Input: `UNIT <units>[term]`

Format: n

<units> 1 = Gauss, 2 = Tesla, 3 = Oersted, 4 = Amp/meter.

Example: **UNIT 2[term] –** Configures the Model 475 to report readings in Tesla.

UNIT? Field Units Query

Input: `UNIT? [term]`

Returned: <units>[term]

Format: n (Refer to command for description)

ZCLEAR Clear Zero Probe Command

Input: `ZCLEAR [term]`

Remarks: Resets the value stored from the ZPROBE command.

ZPROBE Zero Probe Command

Input: `ZPROBE [term]`

Remarks: Initiates the Zero Probe function. Place the probe in zero gauss chamber before issuing this command.